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Advanced Machine Learning
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Exercise Sheet 8
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Objectives of this exercise sheet:

- Exercise 1 offers some insight into fast convolutions, here, using the FFT. It is an implementation task focusing on the inference and the feedforward. The idea is that if you work on this task you will gain a deeper understanding of fast convolutions using the FFT. This homework exercise is needed for Exercise 5 in class, at least give it a try before the exercise takes place.

- Exercise 2 is the mathematical task: to understand the approach of the FFT for $n = 2^k$ it is best to try to recap it for $n = 3^k$. Even if you fail, you will have gained a substantiable knowledge about the FFT. Yes, it is mathematical, but in some sense, also straightforward if you understand the approach for $n = 2^k$.

- Exercise 3 is to show you that even enlarging the problem can lead to a faster algorithm. At the same time, it should be perfectly doable.

- Exercise 4 is like Exercise 1, but this time using `im2col` in place of the FFT. The complicated parts of `im2col` have been treated in the lecture. Like Exercise 1, also this in-class exercise is needed for Exercise 5.

- Exercise 5 is to give you a feeling for the fast convolution algorithms (neglecting here Winograd's fast convolution). It should spark your interest and help you memorize the cases where each variant wins (e.g., small versus large filters).

Remark: In order to use our library it has to be registered with `pip` for each exercise sheet. You can use the script `uninstall_amllib.py` to make sure the library version from the last exercise sheet is deleted. Then use `install_amllib.py` to register the library version for this exercise sheet. The script `install_amllib` registers the library as editable library, hence all changes are directly accessible. The script `uninstall_amllib.py` reverts the installation.

Homework:

Exercise 1: (4 points) level ●●○ relevance ●●● required for exercise 5
In Exercise 2 on the previous exercise sheet you implemented evaluation and backpropagation of a convolutional layer using the function `convolve2D()` of the SciPy package. Now we want to utilize the FFT approach to calculate the convolution.

a) Complete the methods

- `evaluate_fft(self, a)`: evaluation of a 2D convolutional layer with FFT and IFFT.
- `feedforward_fft(self, a)`: evaluation of a 2D convolutional layer with FFT and IFFT, cache data for the backpropagation.

Use the functions `rfft2` and `irfft2` from `scipy.fft`[1].

b) Test your implementation with `tests/test_fft.py`.

Exercise 2: (4 points) level ●●○ relevance ●●○
Derive the FFT for $n = 3m = 3^k$ by following the approach for $n = 2m = 2^k$ shown in the lecture. You only need to derive the recursive variant. Hint: you need to decompose the DFT for $n = 3m$ into three DFTs of size m.

Exercises (in class):

Exercise 3: (4 points) level ●●○ relevance ●●○

a) Count operations (additions and multiplications) in the following two algorithms to compute the full convolution $\mathbf{y} = \mathbf{x} * \mathbf{f} \in \mathbb{R}^{n+k-1}$ of $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{f} \in \mathbb{R}^k$, $n \geqslant k$:
   (i) directly compute the full convolution,
   (ii) enlarge the dimension of $\mathbf{x}$ and $\mathbf{f}$ to the smallest $2^\ell \geqslant n+k-1$ by appending trailing zeros and use the radix-2 FFT/IFFT (w/o the permutation).

b) Write a Python script that plots for given $n \in \mathbb{N}$ the operation count of both approaches. For which k is FFT better?

Hint: You can use the fact that one full FFT computation needs $\ell 2^\ell$ multiplications and additions for part (ii) of a) (see the python snippet from Lecture 6).

Exercise 4: (4 points) level ●●○ relevance ●●● required for exercise 5
In analogy to Exercise 1 we want to utilize the `im2col` approach to calculate the convolution.

a) Complete the methods
   - `evaluate_im2col(self, a)`: evaluation of a 2D convolutional layer with im2col,
   - `feedforward_im2col(self, a)`: evaluation of a 2D convolutional layer with im2col, cache data for the backpropagation.

   Use the function `im2col` from `utils/im2col_utils.py` to map the input to the corresponding im2col matrix. Use the attribute `cache` of `Conv2DLayer` to save the im2col matrix and the reshaped filterbank. Don't forget to reshape the result before applying the activation function.

b) Test your implementation with `tests/test_im2col.py`.

Exercise 5: (3 points) level ●○○ relevance ●●○
Use the file `exercises/exercise5.py` to compare the computation time for the different implementations of the evaluation of a convolutional layer. You might want to change some parameters. What do you oberserve?

[1] More information can be found at https://docs.scipy.org/doc/scipy/tutorial/fft.html.